

Astronomical SNR Calculator

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This form calculates the Signal to Noise Ratio (SNR) for aperture photometry of a point source. The SNR is calculated for filtered or clear observations using the properties of the sensor (CCD), instrument (telescope and filter), and the observation. Use this calculator to plan observations by estimating the effect of exposure time and other parameters. You also can work backward to determine what combination of parameters gives a target SNR. Click [Save] to save parameters, click [Load] to fetch saved parameters. Use [Default] to load reasonable default parameters. Also see the Optimizer tool. [Submit Comments]

Load Save Clear Default

Camera

Pixel Size	Binning	Readout Noise	Gain
9 microns	1	16 electrons	1.5 e- / ADU

Temp	T doubling	Tref	Dark Current
-10 ° C	6.3 ° C	0 ° C	0.19778 pA / cm^2 @ Tref

Instrument

Focal Length	Diameter	Secondary	Efficiency
3910 mm	355 mm	100 diameter, mm	100 percent

Bandpass

Name	Filter Factor	QE	Extinction	Airmass
Johnson U	90 percent (100%)	70 percent	1 mags @ zenith	1.1 zenith = 1.0

Observations

Object Mag	Exp Time	Seeing	Aperture	Sky Mag
7.5 (bandpass)	180 seconds	3.5 FWHM, arcsec	4 radius, binned pix	18.5 mag / arcsec^2

Calculate All results are in units of binned pixels.

Results

SNR	Sigma(m)
3385.846	0.000 +/- mag

Calculations

Dark Current	Base Noise	Image Scale	Object Flux
0.333 e- / s / pix	17.776 e- / pix	0.475 arcsec / pix	42612.315 ADU / s

Sky Flux	Total BG Flux	Total BG SNR	Sky / BG
1.889 ADU / s / pix	2.111 ADU / s / pix	23.873	0.895 Noise Ratio

Aperture	Aperture	Aperture Area	Ap Integral
1.085 diam (FWHM)	3.798 diam (arcsec)	11.331 arcsec^2	55.755 percent

Notes

Camera

Pixel Size
The fundamental width of the detector photo-site, assumed to be square. To calculate for a binned image, change the *Binning* field but do not change the fundamental *Pixel Size*. For example, suppose your CCD has a 9 micron pixel. To use a 2x2 binned image with effective 18 micron pixels, set *Pixel Size*=9 and *Binning*=2. All calculations are made for binned pixels.

Binning
The number of detector pixels per image pixel. This assumes the detector is binned n x n, where n = 1, 2, etc. See the *Pixel Size* parameter.

Readnoise
The detector readout noise is the intrinsic rms noise in a zero second exposure.

Gain
The number of electrons per ADU (Analog to Digital Unit) or DN (Digital Number) in the image.

Temp
The temperature of the detector.

T doubling
The doubling temperature for dark current approximated as a line in a log-log plot of dark current versus temperature. See the detector's data sheet.

Tref
The reference temperature for the detector *Dark Current*. See the detector's data sheet.

Dark Current
The rate of thermal electron production at a refernece temperature. See the detector's data sheet. The dark current is calculated using a simple doubling rule starting from a known reference dark current at a reference temperature. This is a very good approximation when the sensor temperature is not too far below the reference temperature, say within 5 or 6 doublings. However, getting even further below the reference temperature, the dark current can become so small that it has negligible effect on the calculated SNR.

Instrument

Focal Length
The telescope focal length in mm.

Diameter
The telescope primary diameter in mm.

Secondary
The telescope secondary obstruction in mm.

Efficiency
The percentage transmission of the optical system in the selected bandpass.

Bandpass

Name
The name corresponding to the filter set. This selects the photon flux for a star with the spectrum of Vega and magnitude 0 in the selected bandpass above the atmosphere.

Filter Factor
A response or transmission factor for the bandpass filter relative to the standard bandpass. This should usually be 100% unless using a low transmission filter.

QE
The quantum efficiency of the detector in the named bandpass. See the detector's data sheet.

Extinction
The atmospheric extinction coefficient per unit airmass in the named bandpass. This is site dependent. If set to 0.0, then nominal values will be entered into the field when [Calculate] is clicked.

Airmass
The airmass, or optical path length through the atmosphere. The shortest path is through the zenith, which has the minimal airmass value of 1.0. The airmass of an observation is approximated by sec(z), which is the secant of the zenith angle. If set to 0, the observation is considered above the atmosphere.

Observations

Magnitude
The magnitude of the target star through the bandpass filter. This magnitude must be adjusted to the bandpass, e.g., use a V magnitude for the V bandpass, B magnitude for the B bandpass, etc.

Exp Time
The exposure time in seconds.

Seeing
The FWHM of the point spread function which is a product of the atmospheric smearing and optical abberations.

Sky Mag
The sky brightness per square arcsecond in the named bandpass. Sample values for a very dark site are around 21.8 to 22 in B and 21.3 to 21.5 in the V bandpass.

Computed Values

Aperture
This is the value of the measuring aperture *radius* that you would specify in Mira. Some software packages specify the aperture in terms of diameter, not radius. Be sure you enter the appropriate value. Enter a value using binned pixels that you see in the image. For example, setting *Aperture* = 9 at *Binning*=1 gives the same arcsecond radius as *Aperture* = 4.5 at *Binning*=2.

Aperture Integral
The fraction of the star's light inside the photometry aperture, assuming a circular Gaussian profile and a circular aperture.

Total BG Noise
The total noise in the background underneath the star. This includes readout noise, digitization noise, dark noise, and photon noise from the sky.

Sky / BG

The ratio of sky photon noise to total background noise (which includes sky noise). If this field is 1.0, then the sky photon noise is the only source of noise in the background underneath the star. This result helps you address the question of whether an observation is "sky limited" or "sensor limited".

Reference for equations and theory:
Newberry, M. V. 1991, PASP 103, 122-130.

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